SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 6 **Total Marks:30**

Annexure A

Scenario-Based

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Scenario-Based

1. Smart Healthcare Management System (SDG 3: Good Health and Well-being)

A government hospital plans to implement a Smart Healthcare Management System to improve patient care and reduce waiting time. Patients should be able to register online, schedule appointments, access electronic health records, receive digital prescriptions, and make online payments. Doctors, nurses, pharmacists, and laboratory technicians interact with the system to update medical information. The administration also requires automatic emergency alerts, medicine inventory monitoring, and health analytics dashboards. During system analysis, it is observed that patient information is duplicated across multiple departments, leading to inconsistencies and delayed treatment.

Analyze the functional and non-functional requirements of the system. Identify the actors, classes, relationships, and interactions involved. Recommend appropriate UML diagrams to represent the system architecture and justify your choice. Suggest object-oriented design improvements that eliminate redundancy and improve healthcare service delivery while supporting SDG 3.

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2. Smart Irrigation and Water Resource Management (SDG 6: Clean Water and Sanitation)

A state agriculture department plans to develop an IoT-enabled Smart Irrigation System to reduce water wastage. Soil moisture sensors, weather forecasting services, water pumps, and mobile applications are integrated into the system. Farmers receive irrigation recommendations based on crop type and weather conditions. Administrators monitor reservoir levels and water consumption statistics. During implementation, inconsistent communication between IoT devices and the central server results in incorrect irrigation schedules.

Analyze the communication issues affecting the system. Identify the required classes, interactions, and software components. Recommend suitable UML diagrams to model the system and propose object-oriented design improvements that enhance reliability, scalability, and efficient water utilization in support of SDG 6.

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3. Renewable Energy Monitoring System (SDG 7: Affordable and Clean Energy)

A smart city plans to deploy solar panels and wind turbines connected through an Energy Monitoring System. The application records energy production, battery storage levels, equipment status, and power distribution. Engineers monitor system performance using real-time dashboards, while consumers access energy consumption reports through a mobile application. Performance analysis reveals delays in updating energy statistics due to tightly coupled software modules.

Analyze the architectural limitations of the system. Recommend suitable UML diagrams to represent object relationships, interactions, software components, and deployment architecture. Suggest object-oriented design strategies that improve modularity, maintainability, and efficient renewable energy management while contributing to SDG 7.

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4. Smart Public Transportation System (SDG 11: Sustainable Cities and Communities)

A metropolitan transport authority plans to develop a Smart Public Transportation System to encourage the use of buses and metro services. Passengers should be able to track vehicle locations, reserve seats, purchase digital tickets, and receive delay notifications. Traffic monitoring sensors provide real-time congestion information to optimize routes. During testing, passengers receive incorrect arrival times because data synchronization between GPS devices and servers is inconsistent.

Analyze the system requirements and identify the root cause of synchronization failures. Recommend appropriate UML diagrams to represent system behavior, interactions, and deployment architecture. Suggest object-oriented design improvements that improve service reliability and support sustainable urban transportation under SDG 11.

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5. Smart Waste Management System (SDG 11: Sustainable Cities and Communities)

A municipal corporation plans to implement a Smart Waste Management System using IoT-enabled waste bins. Sensors monitor fill levels, collection vehicles receive optimized routes, and administrators monitor waste collection efficiency. Citizens can report overflowing bins through a mobile application. During deployment, duplicate waste collection requests are generated because multiple sensors transmit inconsistent data simultaneously.

Analyze the system architecture and identify the software design issues responsible for duplicate requests. Recommend suitable UML diagrams to represent system components, workflows, and communication among objects. Suggest object-oriented solutions to improve data consistency and operational efficiency while promoting sustainable city management under SDG 11.

RUBRICS FOR CALCULATION

Scenario-Based

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand